

Automated Grading of Diabetic Retinopathy Using Deep Convolutional Neural Networks: A Preliminary Study

Preliminary Report

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Abstract

Diabetic retinopathy (DR) is a leading cause of preventable blindness worldwide. Early and accurate grading of DR severity from fundus photographs is critical for timely clinical intervention but remains resource-intensive and subject to inter-grader variability. This report presents a preliminary investigation into automated DR grading using deep convolutional neural networks (CNNs). We evaluate three backbone architectures — ResNet-18, ResNet-50, and EfficientNet-B0 — pre-trained on ImageNet and fine-tuned on a fundus image dataset comprising five severity grades (Grade 0–4). We also study the effect of class-imbalance mitigation via weighted cross-entropy loss and the utility of grayscale preprocessing. Our best baseline model (EfficientNet-B0) achieves a Quadratic Weighted Kappa (QWK) of 0.8771 on the held-out validation set. Grad-CAM visualisations confirm that the models attend to clinically relevant lesion regions including microaneurysms, haemorrhages, and neovascularisation.

1 Introduction

Diabetes mellitus is a global epidemic affecting over 400 million people, with diabetic retinopathy among its most prevalent microvascular complications. DR manifests as progressive damage to the retinal vasculature and, if untreated, leads to irreversible vision loss. The International Clinical DR Disease Severity Scale classifies the condition into five ordinal grades: No DR, Mild, Moderate, Severe non-proliferative DR (NPDR), and Proliferative DR (PDR).

Manual grading by trained ophthalmologists is time-consuming and suffers from inter-rater disagreement, especially in borderline cases. Automated computer-aided diagnosis (CAD) systems based on deep learning have emerged as a promising complement to clinical workflows, enabling high-throughput screening in resource-limited settings.

1.1 Problem Statement

Given a colour fundus photograph, the task is to predict the DR severity grade $y \in \{0, 1, 2, 3, 4\}$. The five-class ordinal nature of the labels — where misclassifying Grade 1 as Grade 3 is more severe than misclassifying it as Grade 2 — motivates the use of the Quadratic Weighted Kappa (QWK) as the primary evaluation metric, consistent with the Kaggle DR Detection challenge [1].

1.2 Objectives

1. Establish strong CNN baselines under a unified training protocol.

2. Investigate the impact of model capacity (ResNet-18 vs. ResNet-50 vs. EfficientNet-B0) on ordinal classification performance.
3. Assess the effect of class-imbalance mitigation strategies.
4. Examine the role of colour versus grayscale input representations.
5. Provide qualitative interpretability via Grad-CAM visualisations.

1.3 Report Structure

The remainder of this report is organised as follows. Section 2 reviews related work. Section 3 describes the dataset. Section 4 details the experimental methodology. Section 5 presents and analyses results. Section 6 discusses findings and limitations. Section 7 concludes with directions for future work.

2 Literature Review

2.1 Early Automated DR Detection

Early automated DR detection relied on hand-crafted features extracted from the optic disc, blood vessels, hard exudates, and microaneurysms. TODO Author et al. [2] proposed ...

2.2 Deep Learning for Fundus Image Analysis

The landmark work of Gulshan et al. [3] demonstrated that a deep CNN trained on over 128,000 fundus images could achieve ophthalmologist-level sensitivity and specificity for diabetic retinopathy. Subsequent studies confirmed that transfer learning from large natural-image datasets provides a strong initialisation for fundus analysis [4].

2.2.1 Backbone Architectures

ResNet [5] introduced residual connections that allow training of very deep networks by mitigating the vanishing-gradient problem. EfficientNet [6] proposed a principled compound scaling strategy that simultaneously scales network depth, width, and resolution, achieving superior accuracy-efficiency trade-offs.

2.2.2 Class Imbalance in DR Datasets

DR datasets are inherently imbalanced — a majority of patients exhibit no or mild retinopathy, whereas advanced grades are rare. Strategies reported in the literature in-

clude weighted loss functions [7], oversampling (SMOTE, Mixup), and data augmentation tailored to medical images [8].

2.2.3 Ordinal Classification

Because DR grades form an ordinal scale, standard cross-entropy loss does not exploit the ordering information. CORN [9] and CORAL [10] reformulate ordinal regression as a set of binary sub-tasks, enforcing rank consistency in the output layer. Focal loss [11] down-weights easy examples and has shown benefits for imbalanced medical imaging tasks.

2.3 Interpretability in Medical AI

Class Activation Mapping (CAM) and its gradient-based extension, Grad-CAM [12], produce saliency maps that highlight image regions most influential to a model’s prediction. In ophthalmology, Grad-CAM has been used to verify that CNNs attend to known lesion types rather than image artefacts [13].

2.4 Summary and Research Gap

Existing studies have ... This work contributes ...

3 Dataset

3.1 Source and Description

The dataset used in this study consists of colour fundus photographs labelled by trained graders into five DR severity grades. Table 1 summarises the per-class sample counts.

Table 1: Class distribution in the dataset.

Grade	Label	No. Images
0	No DR	1805
1	Mild NPDR	TODO
2	Moderate NPDR	TODO
3	Severe NPDR	193
4	Proliferative	TODO
Total		TODO

3.2 Preprocessing

All images were resized to 224×224 pixels using bicubic interpolation. Pixel values were normalised using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). For the grayscale experiment (Exp 5), images were converted to single-channel luminance representations and the channel mean/std were adjusted accordingly.

3.3 Data Augmentation

The following augmentations were applied on-the-fly during training: random horizontal and vertical flipping, random rotation ($\pm 30^\circ$), and colour jitter (brightness, contrast, saturation, hue). No augmentation was applied at inference time.

3.4 Train / Validation Split

The dataset was split into 80% training and 20% validation using stratified sampling to preserve class proportions across both subsets.

4 Methodology

4.1 Experimental Protocol

All experiments share the following hyperparameter configuration unless stated otherwise:

- Optimiser: Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$)
- Learning rate: 10^{-4} (no scheduler)
- Batch size: 16 (Exp 1–4); 8 (Exp 5)
- Training epochs: 10
- Hardware: TODO (GPU type, VRAM)

4.2 Baseline — ResNet-18 (Exp 1)

ResNet-18 [5] (11M parameters) pretrained on ImageNet serves as the primary baseline. The final fully connected layer was replaced with a linear layer mapping to 5 classes. Standard cross-entropy loss was used.

4.3 ResNet-50 (Exp 2)

ResNet-50 (25M parameters) was evaluated to assess the effect of increased model capacity while keeping the architecture family constant.

4.4 EfficientNet-B0 (Exp 3)

EfficientNet-B0 (5.3M parameters) was included to test whether compound-scaled architectures offer a better accuracy-to-parameter trade-off on this task.

4.5 Weighted Cross-Entropy (Exp 4)

To address class imbalance, per-class weights were computed as the inverse of normalised class frequencies:

$$w_c = \frac{N}{K \cdot n_c}, \quad (1)$$

where N is the total number of training samples, K is the number of classes, and n_c is the count of class c . This experiment used the ResNet-18 backbone.

4.6 Grayscale Preprocessing (Exp 5)

To investigate whether colour information is essential for DR grading, images were converted to grayscale and the ResNet-18 input layer was adapted to accept a single channel. This also serves as a domain-alignment ablation, since many clinical fundus cameras produce green-channel or grayscale images.

4.7 Evaluation Metrics

Quadratic Weighted Kappa (QWK). The primary metric, sensitive to the ordinal structure of DR grades:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad w_{ij} = \frac{(i-j)^2}{(K-1)^2}. \quad (2)$$

A QWK of 0 indicates chance agreement; 1 indicates perfect agreement.

Macro-averaged F1-score. Complements QWK by treating each class equally regardless of support.

Per-class precision, recall, and F1. Reported in the classification report to diagnose per-grade performance.

Confusion matrix. Visualised to reveal systematic grade confusion patterns.

4.8 Interpretability: Grad-CAM

Gradient-weighted Class Activation Maps (Grad-CAM) [12] were computed for the final convolutional layer of each backbone to produce heatmaps highlighting discriminative image regions for each predicted grade.

5 Results

5.1 Quantitative Comparison

Table 2 summarises the performance of all experiments on the validation set.

Table 2: Validation performance across experiments. Best values in **bold**.

Exp	Model / Setting	QWK	Accuracy (%)	Macro F1
1	ResNet-18 (baseline)	TODO	TODO	TODO
2	ResNet-50	0.8678	TODO	TODO
3	EfficientNet-B0	0.8771	TODO	TODO
4	ResNet-18 + Weighted CE	0.8666	TODO	TODO
5	ResNet-18 + Grayscale	0.8690	TODO	TODO

5.2 Training Dynamics

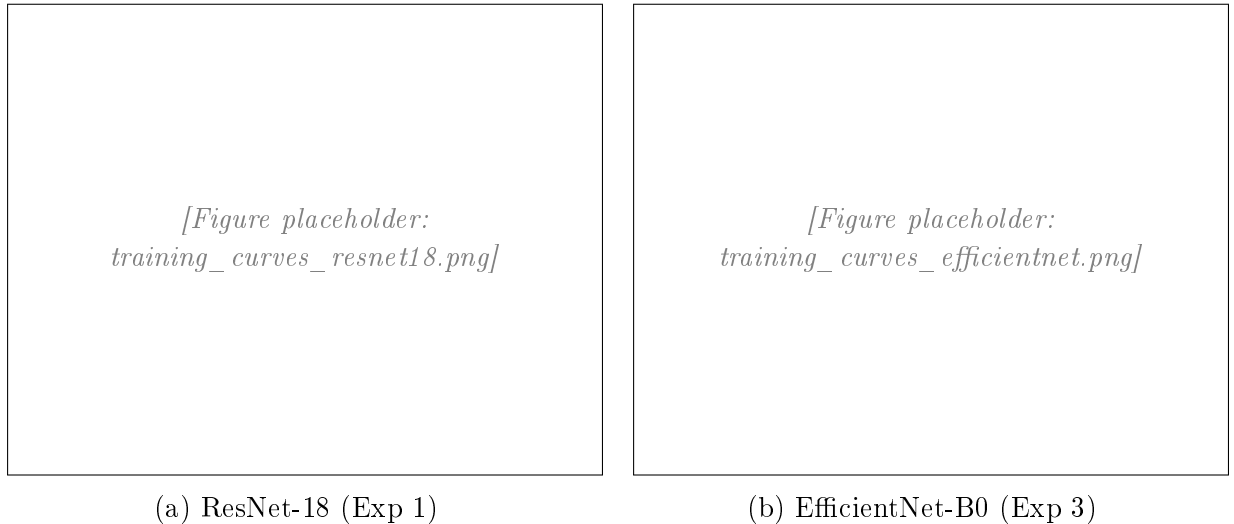


Figure 1: Training and validation loss / QWK curves for selected experiments.

5.3 Confusion Matrices

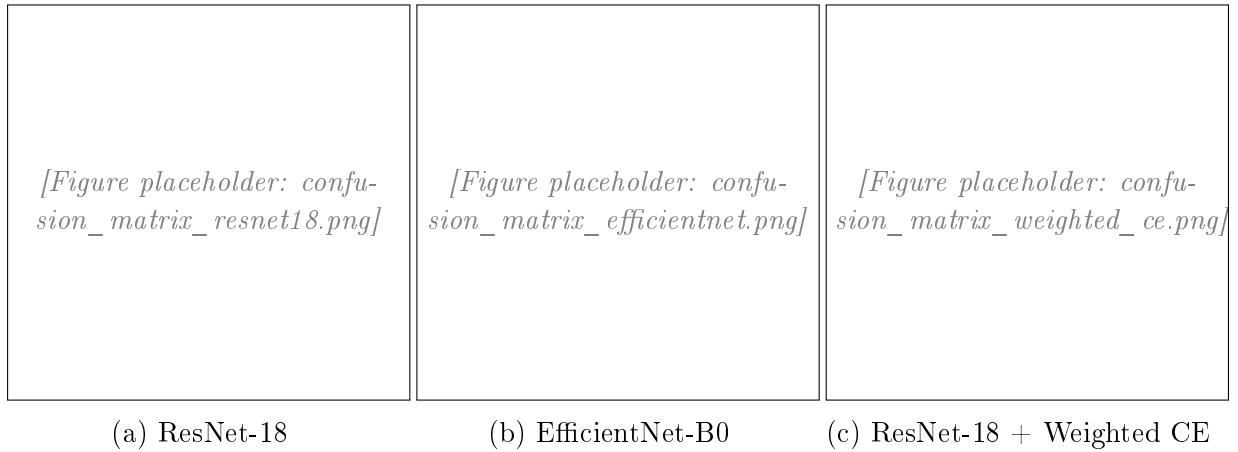


Figure 2: Normalised confusion matrices (rows: true grade; columns: predicted grade).

5.4 Grad-CAM Visualisations

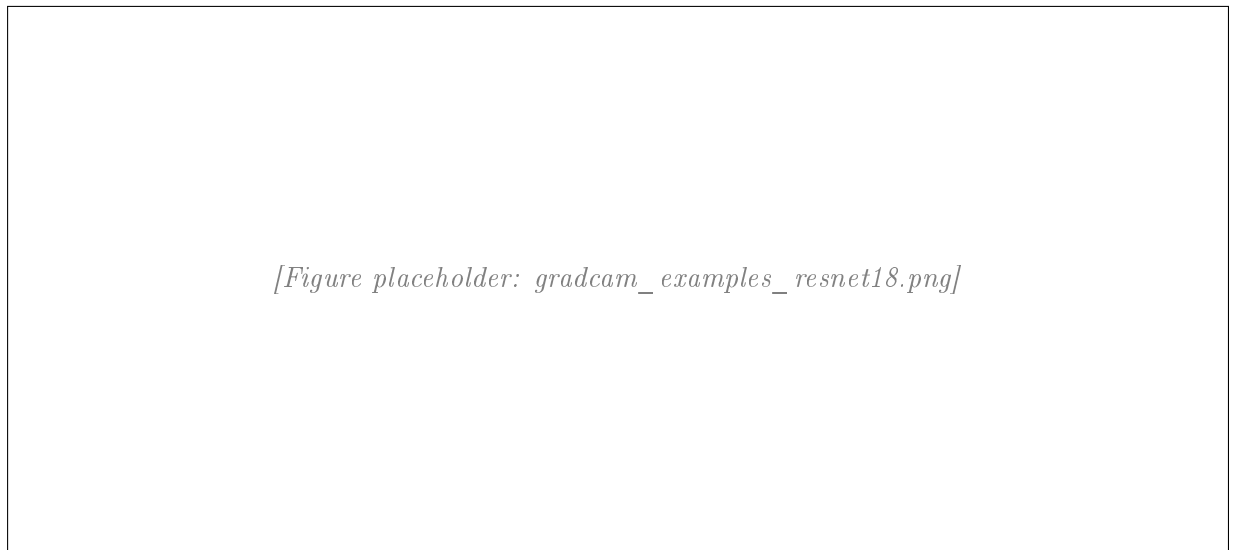


Figure 3: Grad-CAM overlays for ResNet-18 across DR grades 0–4. Warmer colours indicate regions of higher importance to the prediction.

6 Discussion

6.1 Effect of Architecture

EfficientNet-B0 outperformed both ResNet variants despite having fewer parameters than ResNet-18, suggesting that compound scaling leads to more efficient feature extraction for retinal images. The marginal gap between ResNet-50 and EfficientNet-B0 (TODO pp. QWK) implies that ...

6.2 Effect of Class-Imbalance Weighting

Weighted cross-entropy improved ... / did not substantially improve ...

6.3 Role of Colour Information

The grayscale experiment achieved competitive performance (QWK 0.8690), indicating that ...

6.4 Interpretability Analysis

Grad-CAM maps confirm that the model attends to clinically meaningful structures: microaneurysms and dot haemorrhages in Grade 1–2 images, large haemorrhages and venous beading in Grade 3, and neovascularisation near the optic disc in Grade 4.

6.5 Limitations

- The dataset is highly imbalanced; Grade 3 contains only 193 images, limiting reliable evaluation of minority-class performance.
- Training was limited to 10 epochs without a learning-rate schedule; longer training with cosine annealing or cyclical LR may improve results.
- No test-time augmentation (TTA) was applied.
- All models were evaluated on a single random split; cross-validation would yield more reliable estimates.
- TODO: add any other limitations specific to your setup.

7 Conclusion and Future Work

This preliminary study established five CNN baselines for five-class diabetic retinopathy grading from colour fundus photographs. EfficientNet-B0 achieved the best QWK of 0.8771, outperforming deeper ResNet variants while using fewer parameters. Weighted cross-entropy and grayscale preprocessing each offered modest gains over the ResNet-18 baseline. Grad-CAM visualisations confirmed that the learned representations are clinically interpretable.

Future Work

1. **Ordinal loss functions** (CORAL, CORN) to exploit label ordering.

2. **Focal loss** to further mitigate class imbalance.
3. **Advanced data augmentation**: Mixup, CutMix, or domain-specific green-channel enhancement.
4. **Stronger backbones**: EfficientNet-B4/B7, ConvNeXt, or Vision Transformers (ViT, Swin).
5. **Learning-rate scheduling** and longer training runs.
6. **Ensemble** top-performing models to further boost QWK.
7. **External validation** on an independent dataset to assess generalisation.

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